Gold Price Direction Prediction

Index Signals vs. WSJ Text Signals

Group 10

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1. Research Question

Imagine you are a gold investor.

For predicting next-month gold direction, which is more reliable for you:

- index-only signals (UUP and VIX), or
- news-text signals (WSJ headlines)?

2. Research Design

- Goal: Predict next-month GLD direction (up/down).
- Models:
  a. Index-only logistic regression (UUP, VIX)
  b. Text-only logistic regression (WSJ headlines → topics)
- Split: Chronological train / validation / test.
- Main decision metric: MCC (balanced directional reliability).

3.1 Index Model

Data Source: yfinance API

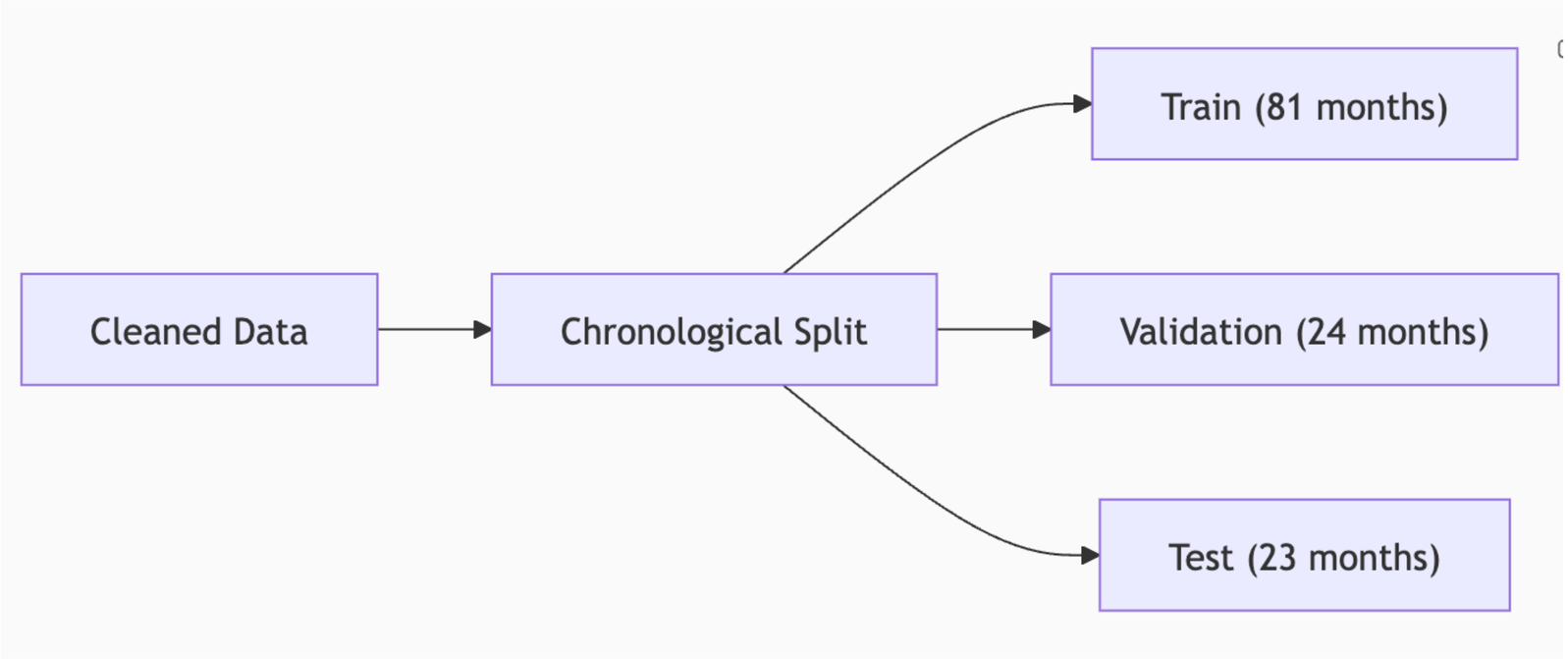
From **March 1, 2007 to December 28, 2017**

Major indexes used:

UUP: An ETF that tracks the performance of the US dollar against a basket of major world currencies. A rising UUP indicates a strengthening US dollar.

VIX: Often called the "fear index", it measures the market's expectation of 30-day volatility for the S&P 500. Higher VIX values indicate increased market uncertainty, fear, and risk aversion among investors.

Index Model Training Pipeline: Logistic Regression



2007-04-> 2013-12

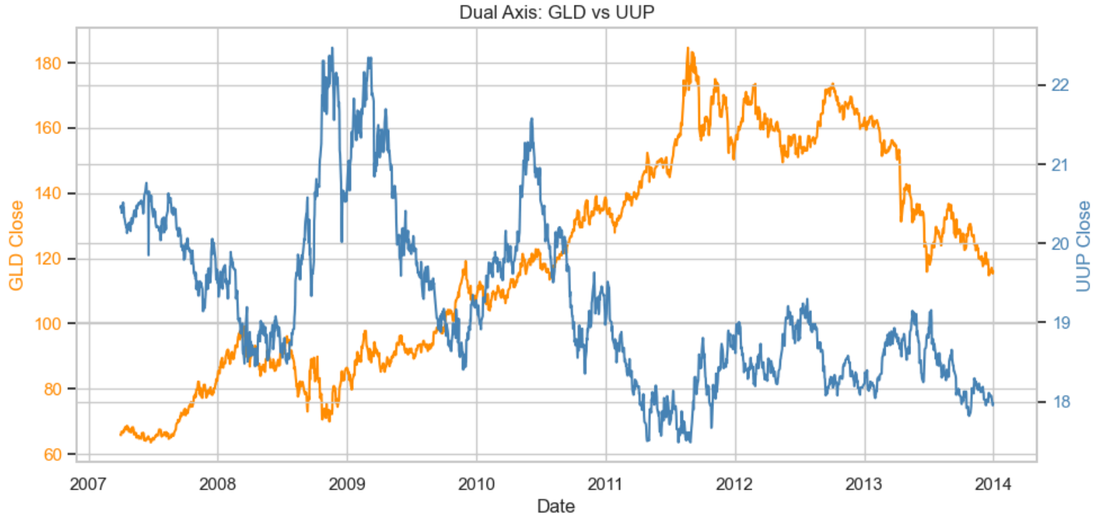
2014-01 -> 2015-12

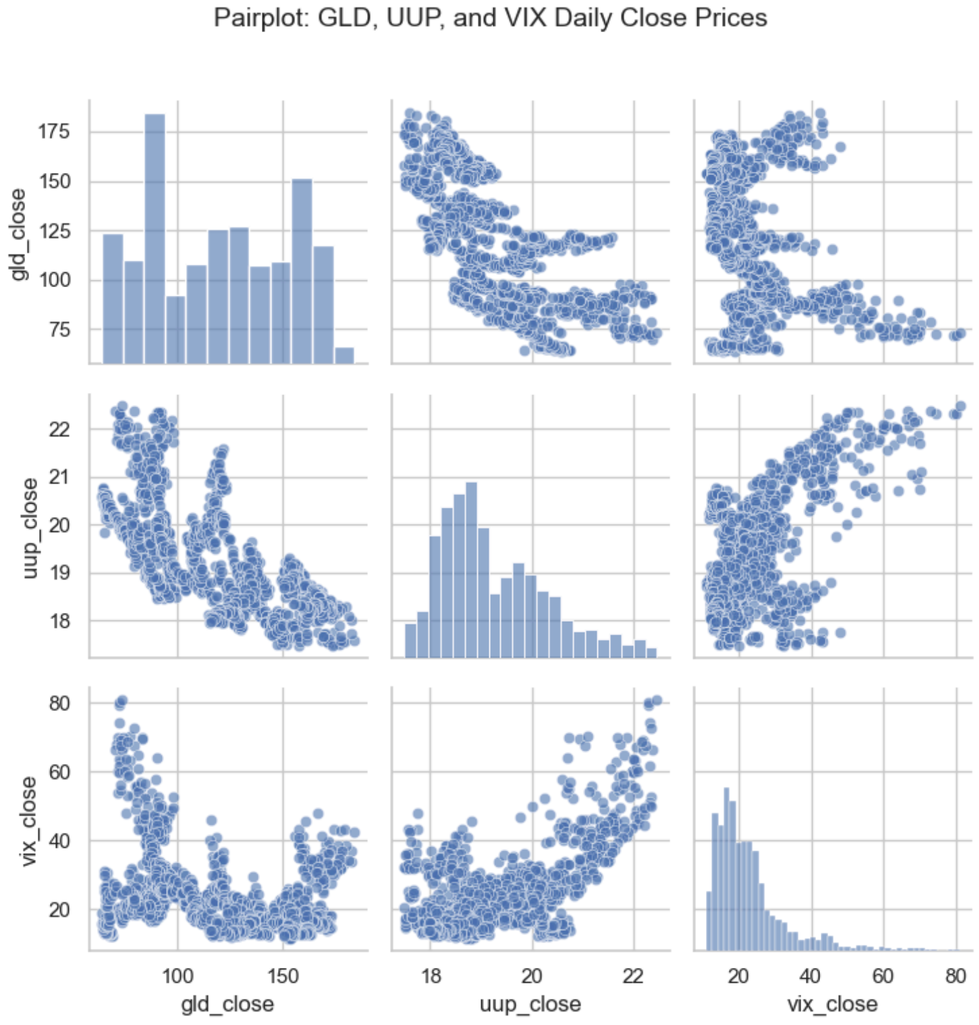
2016-01-> 2017-12

x = UUP & VIX
y = GLD price direction (1 menas up, 0 means down)

EDA on Training Set

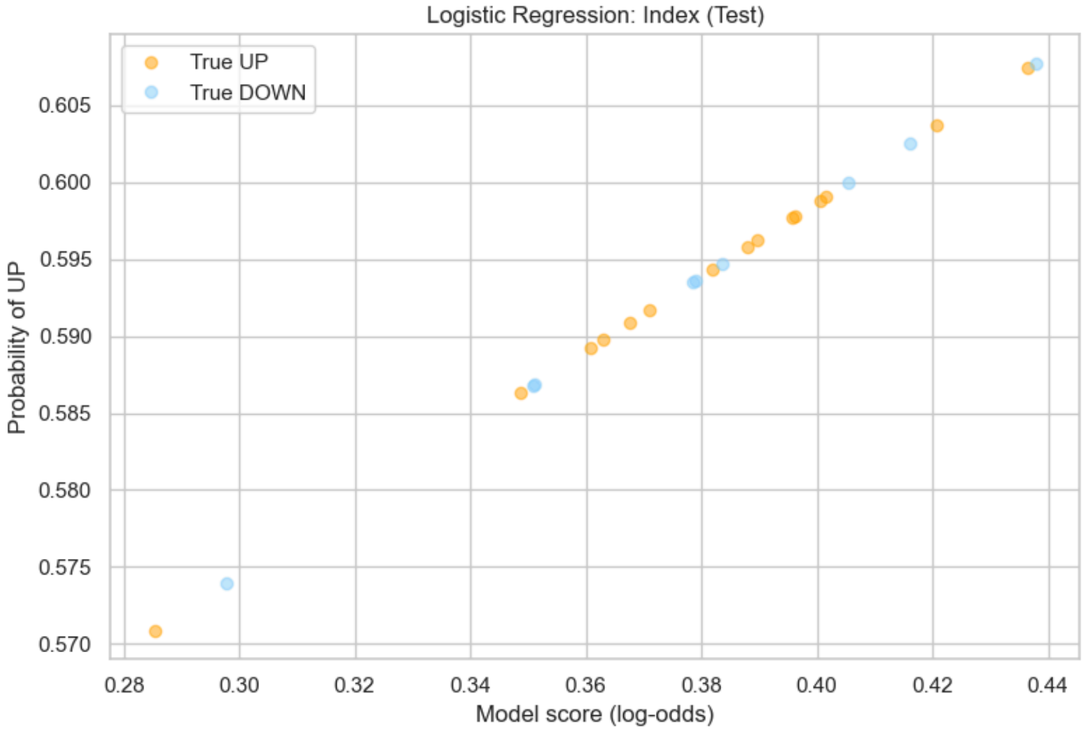
Pairplots and line charts: a negative correlation between GLD and UUP, GLD and VIX.
Correlation: ρ(GLD vs. UUP) = -0.7018, ρ(GLD vs. VIX) = -0.3164





Training and Test Result

Model A: learns little from indexes (MCC<0) (worse than guessing randomly!)



Signal	Split	Threshold	Accuracy	Precision	Recall	F1	ROC-AUC	MCC
index	validation	0.6	0.6667	0.6	0.6	0.6	0.5786	0.3143
index	test	0.6	0.3333	0.4	0.1333	0.2	0.5185	-0.2384

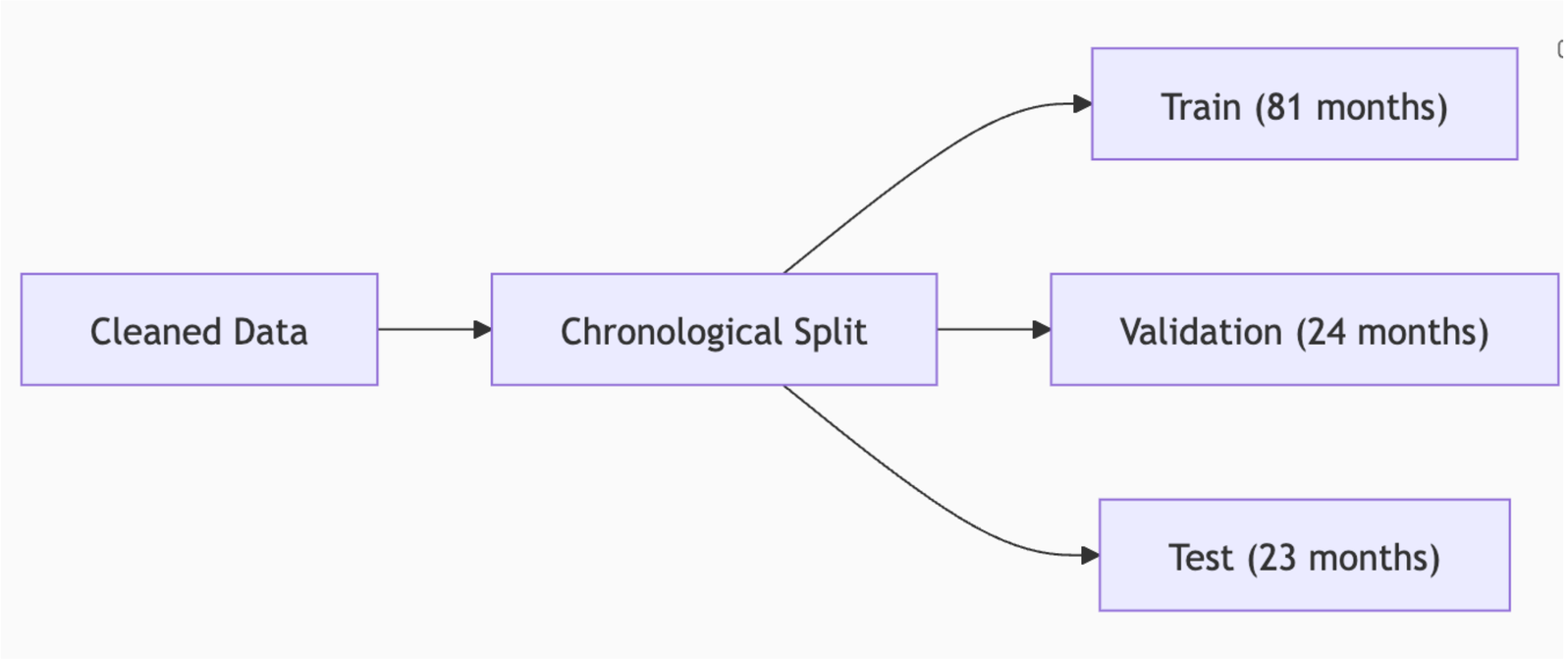
3.2 Text Model

Data Source: WSJ Headlines

From **March 1, 2007 to December 28, 2017**, the WSJ headlines dataset contains **3,250 records**, covering **130 months and 1,925 unique dates**, with about 25 headlines per month on average. The time coverage is continuous enough for monthly aggregation and time-series prediction.

accession_number	display_date	headline
20070301000115	2007-03-01 07:17:21.583	Corporate Focus: Road Could Get Worse for Detroit --- New Fuel Housing Pressures on Buyers May...
20070302000119	2007-03-02 07:15:52.329	Corporate Focus: E.On's Power Play Goes Astray --- Expansion Options Dwindle as Enel's Move Hampers...
20070305000086	2007-03-05 07:12:52.383	Leading the News: Alltel Is Calling Potential Buyers --- List Includes Big Telecoms And Private-Equity Firms...

Text Model Training Pipeline: Logistic Regression



x = topics from the headlines
y = GLD price direction (1 menas up, 0 means down)

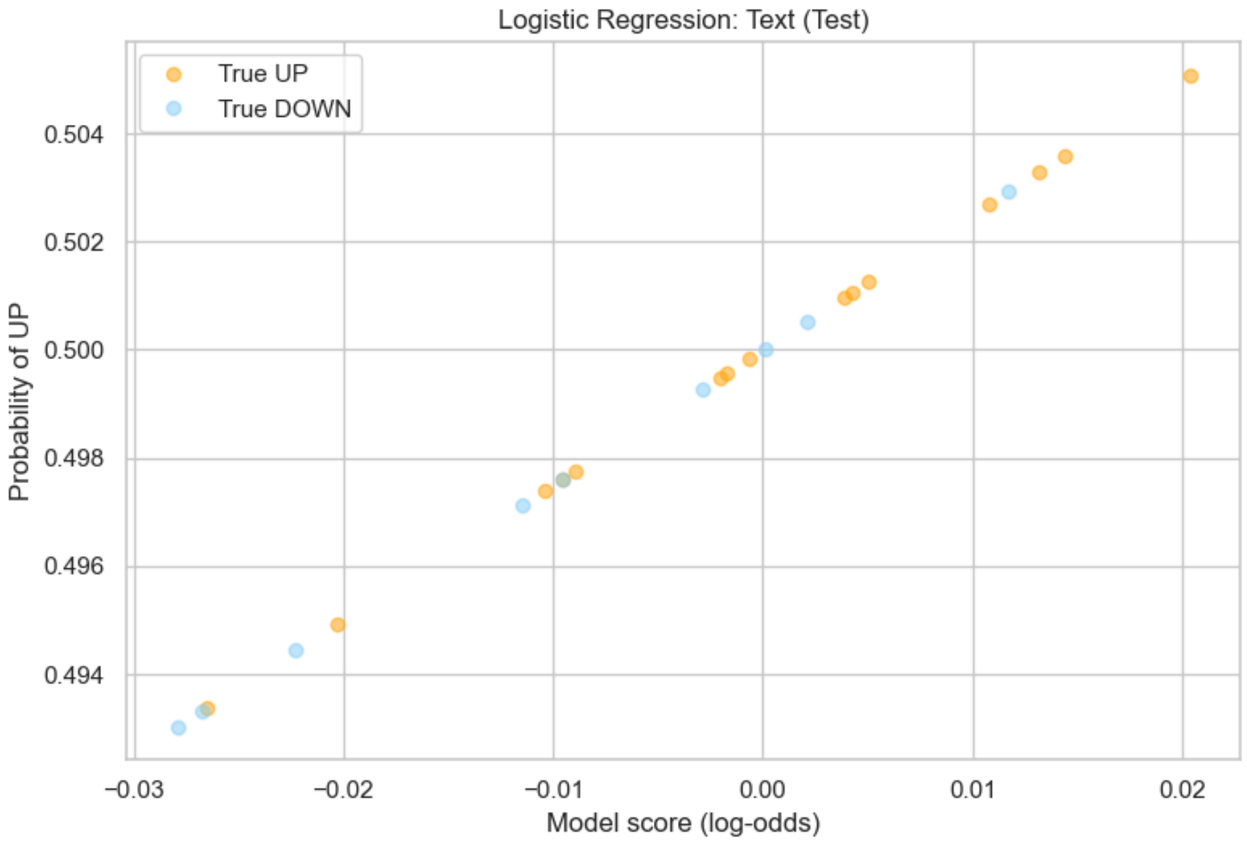
EDA on Training Set

Word Cloud: Cleaned Headlines in Train Set

Aggregation into 12 Topics

topic_label	top_words
topic_0_plan_day_production	plan, day, production, chip, candidate, launch, consumer, cut, focus, rebel, airline, oil
topic_1_case_property_pay	case, property, pay, unit, tax, stake, sign, chief, election, rating, move, party
topic_2_trade_group_security	trade, group, security, drug, term, lender, president, probe, system, terror, pact, age
topic_3_crisis_wall_focus	crisis, wall, focus, policy, plan, change, game, shift, line, program, election, city
topic_4_bid_cash_poll	bid, cash, poll, marketing, fight, technology, bill, suit, end, meeting, name, battle
topic_5_leader_loss_growth	leader, loss, growth, push, wsj, com, technology, power, gop, japan, party, service
topic_6_credit_investor_currency	credit, investor, currency, dollar, oil, debt, bond, city, capital, rate, time, price
topic_7_home_test_rule	home, test, rule, ipo, technology, bond, gas, issue, outlook, plan, way, smartmoney
topic_8_technology_gain_share	technology, gain, share, health, war, asset, court, phone, investor, return, price, strategy
topic_9_profit_china_sale	profit, china, sale, job, cut, outlook, track, energy, offer, growth, cost, rise
topic_10_risk_option_demand	risk, option, demand, analysis, revenue, trader, fee, bofa, jump, investor, action, focus
topic_11_talk_fall_people	talk, fall, people, campaign, aid, post, commodity, auto, help, event, official, plan

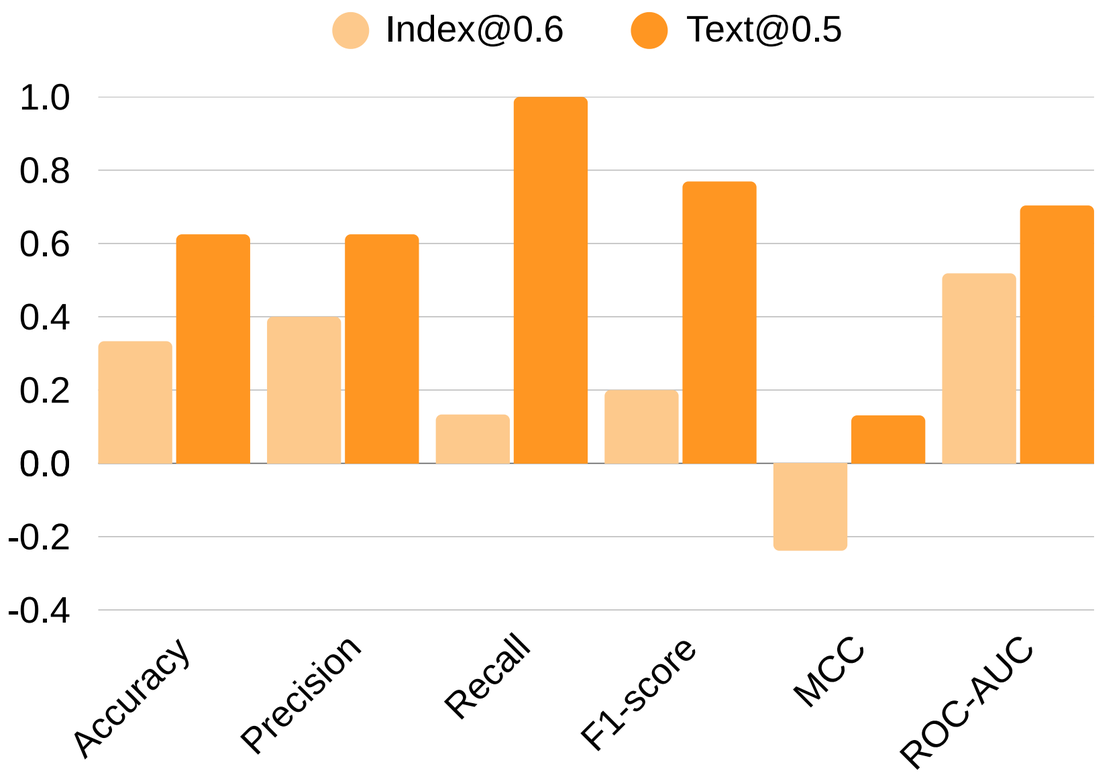
Training and Test Result



Signal	Split	Threshold	Accuracy	Precision	Recall	F1	ROC-AUC	MCC
text	validation	0.5	0.5833	0.5	0.7	0.5833	0.5857	0.2
text	test	0.5	0.5417	0.7	0.4667	0.56	0.7037	0.1309

3.3 Test Result Comparison

Result on Test Set



Text model outperforms the Index Model.

Why?

*MCC, or Matthews Correlation Coefficient, is a balanced classification metric that uses all four outcomes: true positives, true negatives, false positives, and false negatives.

4. Conclusion

Why this can happen:

1. Text does capture some incremental information.
2. Gold-index relationships are time-varying.

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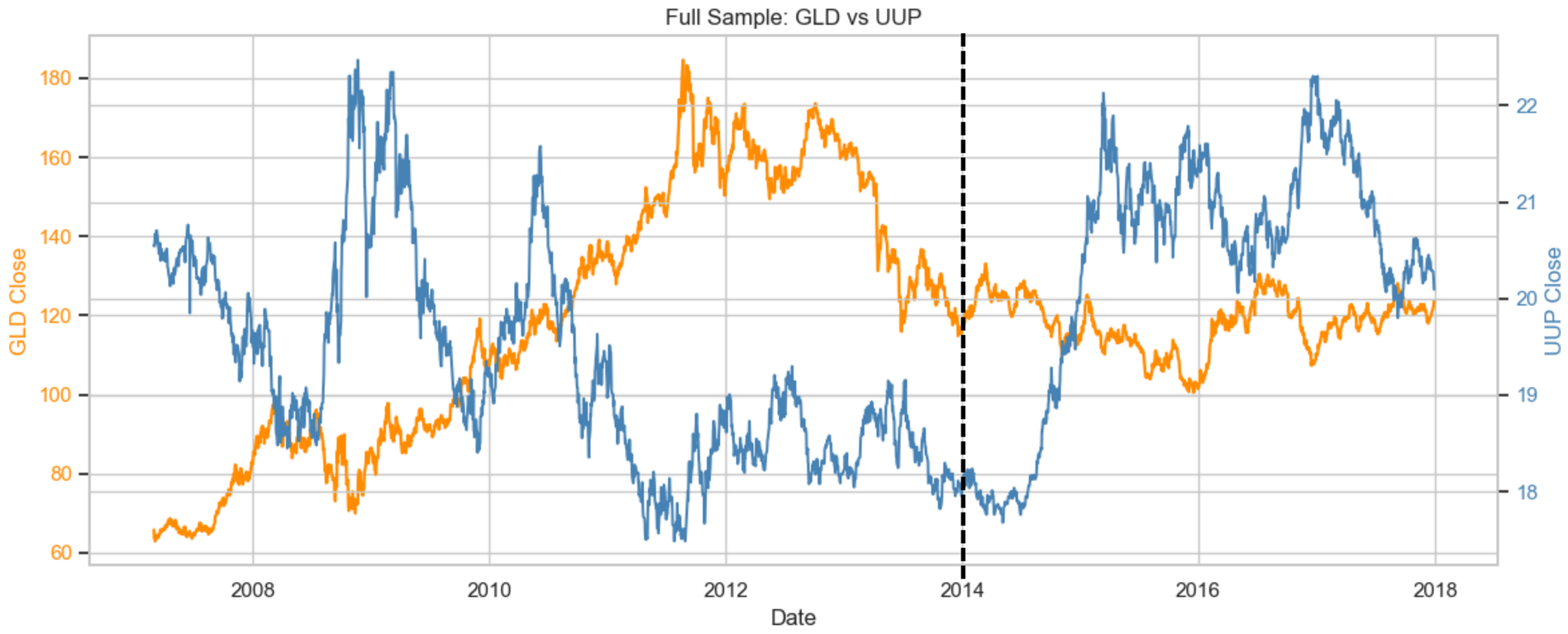


4. Conclusion

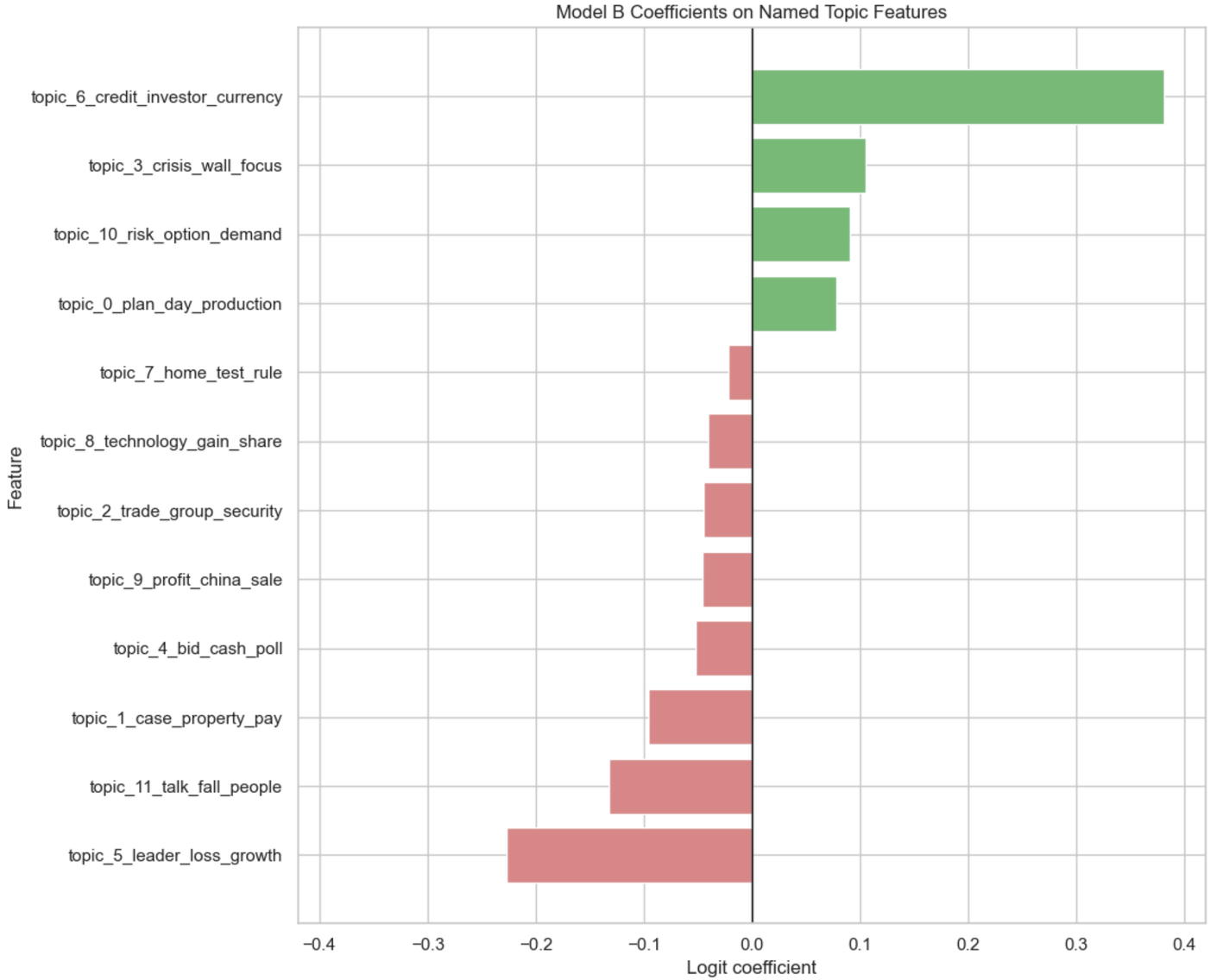
Why this can happen:

1. Text does capture some incremental information.
2. Gold-index relationships are time-varying.
3. A very low signal-to-noise ratio.

Q & A

APPENDIX

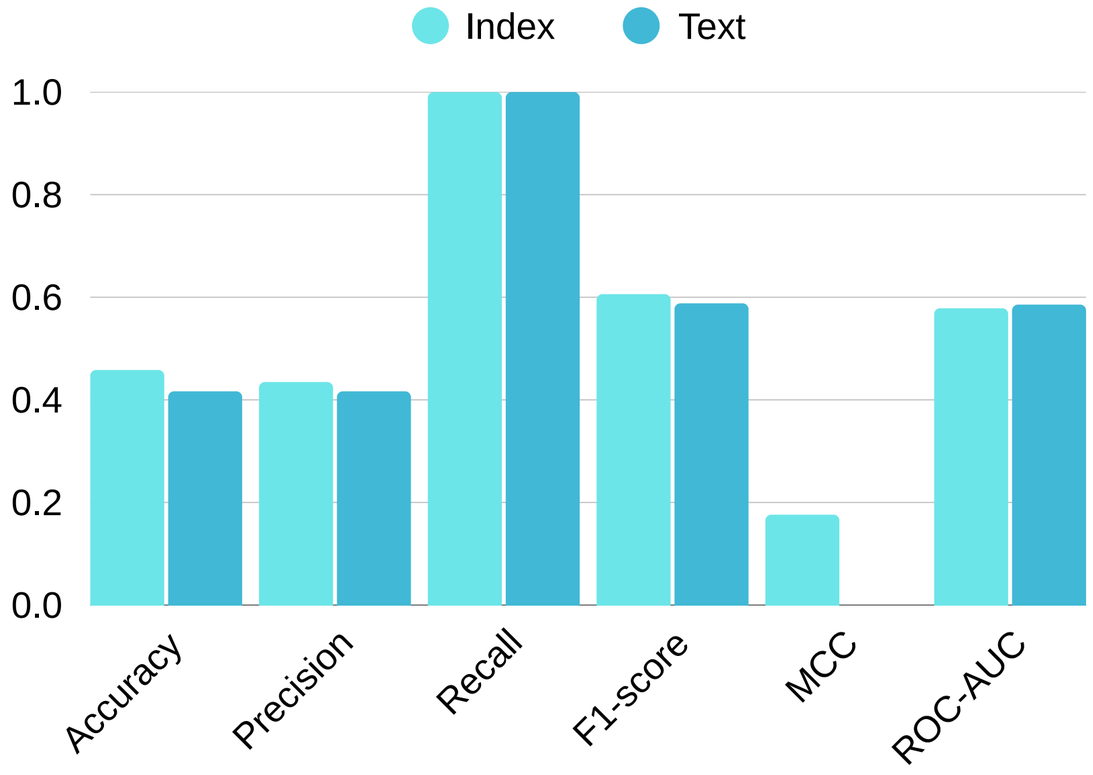
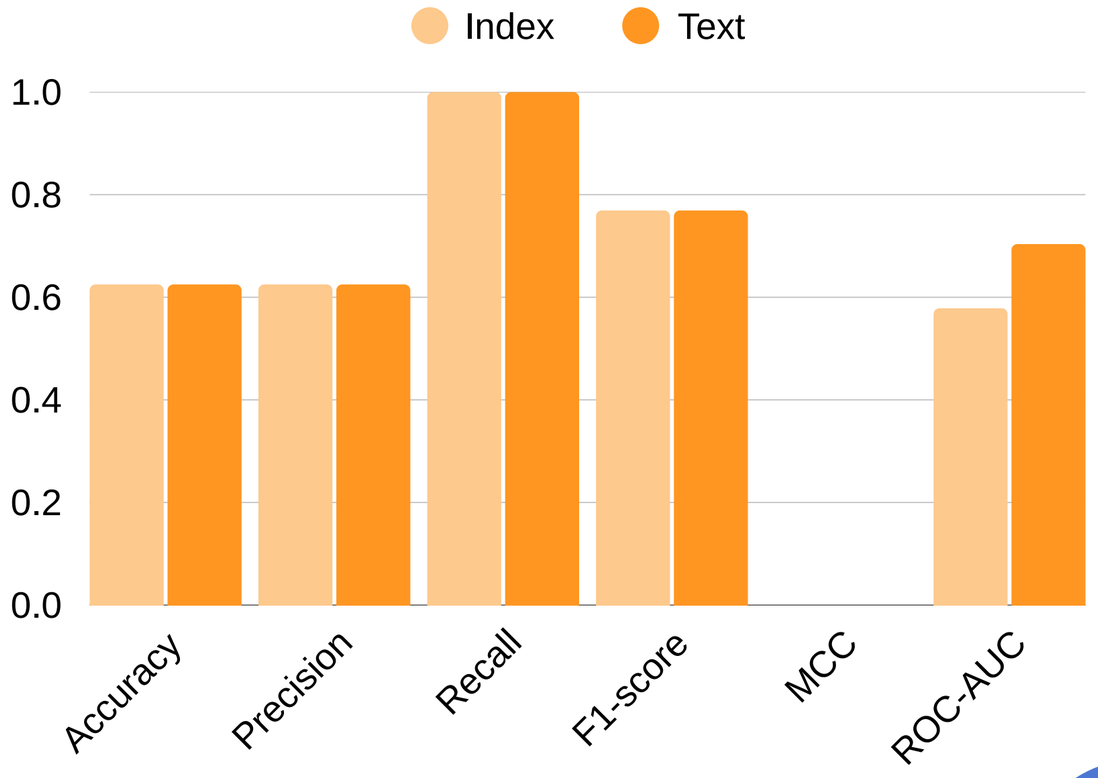
Coefficients on Each Topic



- more noise than signal
- hard to interpret

3.3 Test Result Comparison (Best F1)

Result on Validation Set



Result on Test Set